

BALLYPATRICK FOREST, United KingdomWMO: **039160**Lat: **55.1806N**Lon: **6.1547W**Elev: **512**StdP: **14.43**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	31.5	23.9	18.4	34.0	26.0	20.3	34.6	33.1	41.9	30.1	41.4	5.9	210	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	69.1	61.9	66.4	60.4	64.1	59.0	63.2	67.3	61.5	65.1	59.9	63.2	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	82.9	65.4	59.7	78.0	63.4	58.3	74.0	61.7	28.9	67.3	27.6	65.3	26.6	63.3	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 27.2	23.6	20.8		25.9	74.3	2.5	2.4	24.1	76.0	22.7	77.5	21.3	78.8	19.5	80.6	
(5)			DB	25.3	66.2	2.5	1.6	23.5	67.4	22.0	68.3	20.6	69.2	18.8	70.4	(p)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	40.5	40.7	42.2	45.1	49.4	53.6	56.5	56.5	54.1	49.7	44.6	41.4
	DBStd	7.35	4.41	4.59	4.56	4.54	4.70	4.25	3.92	3.39	3.73	4.19	4.47	4.93
	HDD50	1539	293	262	245	160	69	11	0	0	6	57	169	267
	HDD65	6245	758	680	708	598	485	341	265	264	327	475	613	731
	CDD50	774	0	1	3	12	49	119	202	202	130	47	7	2
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	8	0	0	0	0	0	3	4	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	9.7	11.4	11.2	10.9	9.8	9.3	8.4	7.7	8.0	8.6	10.0	10.2	11.2
	PrecAvg	50.0	5.1	3.9	3.6	3.3	3.1	3.4	3.6	3.9	3.9	5.4	5.5	5.3
	PrecMax	60.0	9.1	8.3	8.4	5.6	8.2	8.6	6.3	10.7	8.0	10.4	9.6	11.2
(15) Precipitation	PrecMin	37.8	1.0	1.2	1.1	0.9	0.6	1.1	1.3	0.8	0.3	2.0	2.9	2.7
	PrecStd	5.9	1.8	1.9	1.7	1.3	1.7	1.8	1.3	2.0	1.8	2.2	1.8	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	53.6	54.7	56.8	61.8	68.8	72.2	73.5	71.3	67.8	61.7	57.4	54.9	
	0.4% MCWB	51.2	50.9	50.2	53.7	58.6	61.4	64.3	64.6	62.1	57.9	55.2	52.8	
	2% DB	51.1	52.1	53.2	57.9	64.4	66.9	69.3	67.7	64.3	59.3	54.9	52.7	
	2% MCWB	49.5	49.5	49.2	52.0	56.6	59.6	62.4	62.2	59.9	56.5	53.0	50.9	
	5% DB	49.3	50.1	51.0	54.7	60.5	63.6	66.1	64.8	61.9	57.6	52.9	50.9	
	5% MCWB	47.8	47.8	47.9	50.3	54.6	57.7	60.8	60.2	58.3	55.2	51.3	49.2	
	10% DB	47.6	48.0	49.2	52.2	57.4	60.8	63.5	62.6	59.9	55.9	51.3	48.9	
	10% MCWB	46.0	45.9	46.5	48.4	52.5	56.3	59.1	58.9	56.7	53.8	49.7	47.2	
	0.4% WB	51.7	51.9	52.5	55.6	60.3	63.3	65.7	65.5	63.4	58.9	55.8	53.5	
	0.4% MCDB	52.7	53.6	55.3	59.7	66.5	69.0	71.2	69.9	66.4	60.8	57.0	54.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	49.8	49.8	50.0	52.9	57.5	60.5	63.3	62.9	60.7	57.1	53.3	51.2	
	2% MCDB	51.1	51.6	52.1	56.8	62.7	65.6	68.0	66.2	63.6	58.6	54.5	52.5	
	5% WB	48.0	48.0	48.4	50.7	55.2	58.5	61.4	61.1	58.9	55.6	51.6	49.4	
	5% MCDB	49.1	49.8	50.5	54.1	59.9	62.6	65.3	64.1	61.2	57.3	52.7	50.7	
	10% WB	46.1	46.0	46.7	48.8	53.0	56.7	59.6	59.4	57.2	54.1	49.9	47.3	
	10% MCDB	47.5	47.9	48.9	51.9	56.7	60.1	62.8	61.9	59.3	55.7	51.1	48.7	
(35) Mean Daily Temperature Range	MDBR	7.2	8.0	9.3	10.1	11.4	10.8	10.3	10.0	9.4	8.3	7.4	7.1	
	5% DB MCDBR	9.2	9.5	11.1	13.2	15.9	15.1	14.5	13.4	11.5	9.5	8.5	9.1	
	5% DB MCWBR	9.1	8.8	8.3	8.9	10.1	9.8	9.4	9.1	8.5	8.0	8.4	9.1	
	5% WB MCDBR	9.2	9.6	10.2	12.2	15.0	14.2	13.5	12.0	10.5	9.0	8.5	9.1	
	5% WB MCWBR	9.2	9.2	8.5	8.9	10.2	10.1	9.4	9.2	8.7	8.2	8.6	9.2	
(40) Clear-Sky Solar Irradiance	taub	0.328	0.340	0.360	0.373	0.376	0.370	0.372	0.358	0.353	0.339	0.322	0.316	
	taud	2.243	2.336	2.335	2.337	2.373	2.421	2.417	2.472	2.480	2.449	2.309	2.203	
	Ebn at Noon	183	232	256	269	275	278	275	272	258	232	187	157	
	Edh at Noon	19	24	30	35	36	34	34	30	26	22	17	16	
	RadAvg	165	382	731	1202	1541	1556	1456	1183	819	452	209	120	
(45) All-Sky Solar Radiation	RadStd	19	36	60	91	136	154	116	93	50	46	21	12	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page